STPM Physics · Semester 3 · Chapter 23

GEOMETRICAL OPTICS

23.1 Spherical Mirrors

DEFINITIONS & LAWS

- **Sign convention used throughout (real-is-positive):** distances of real objects and real images are positive; virtual images (behind the mirror) have negative v; f and R are positive for a concave mirror and negative for a convex mirror.
- The **centre of curvature** C is the centre of the sphere of which the mirror is a part; the **radius of curvature** r is the radius of that sphere.
- The **pole** P is the centre of the reflecting surface of the mirror; the **aperture** is the size (area) of the reflecting surface.
- The **principal axis** is the line through the pole and the centre of curvature; it is normal to the mirror at its pole.
- The **principal focus** of a concave mirror is the point on the principal axis through which rays parallel and close to the axis converge after reflection.
- The **focal length** is the distance from the pole to the principal focus; $f = \frac{R}{2}$ for a spherical mirror of small aperture, where R is the radius of curvature.
- **Linear magnification** is the ratio of the image height to the object height, $m = \frac{h_i}{h_o} = -\frac{v}{u}$; m is negative for an inverted (real) image and positive for an upright (virtual) image.
- A concave mirror forms a real, inverted image for u > f and a virtual, upright, magnified image for u < f; a convex mirror always forms a virtual, upright, diminished image.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mirror equation	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	u = object distance; v = image distance; f = focal length (all in m or cm, same unit)
Focal length and radius	$f = \frac{R}{2}$	f = focal length; R = radius of curvature; f > 0 for concave, f < 0 for convex
Linear magnification	$m = -\frac{v}{u} = \frac{h_i}{h_o}$	m = magnification (no unit); h_i = image height; h_o = object height; $\lvert m\rvert > 1$ magnified, $\lvert m\rvert < 1$ diminished
Image same size as object	$u = v = 2f = R$; $m = -1$	u = object distance; v = image distance; f = focal length; R = radius of curvature; real, inverted image

Quantity / law	Formula	Symbols, units & conditions
Virtual image	$v < 0$; $m = -\frac{v}{u} > 0$	v = image distance; u = object distance; m is positive (upright image)

TIPS TO ANSWER

- State the convention (real-is-positive) before substituting; write f with the correct sign.
- Find v from $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$, then $m = -\frac{v}{u}$; state the nature: real/virtual, upright/inverted, magnified/diminished.
- Give the position as 'v cm in front of / behind the mirror' in words, in agreement with the sign of v.
- In a ray diagram use at least two principal rays (parallel to axis then through F; through C reflecting back on itself), with arrows and labelled F, C, P.
- Comparison questions are a recent trend: be ready to compare the mirror equation with the thin lens formula, and the images formed by concave and convex mirrors.

SECTION A TRAPS

Trap: Using f = R instead of $f = \frac{R}{2}$, or forgetting that R is the distance to the centre of curvature, not the focus.

Trap: Taking f of a convex mirror as positive in the real-is-positive convention; it is negative, so v is negative.

Trap: Reporting m without sign. The sign tells the orientation: negative = inverted.

Trap: Inverting the result: $\frac{1}{v}$ is the reciprocal of v; do not quote $\frac{1}{v}$ as the image distance.

TRICKS

Trick: Limiting case: $u \to \infty$ gives $v = f$; $u = 2f$ gives $v = 2f$ with $m = -1$; $u = f$ gives no image ($v \to \infty$).

Trick: If $u < f$ for a concave mirror, $\frac{1}{f} - \frac{1}{u}$ is negative, so the image is virtual and magnified.

23.2 Refraction at Spherical Surfaces

DEFINITIONS & LAWS

- **Sign convention (real-is-positive):** u is positive for a real object in front of the surface; v is positive for a real image formed on the side where light emerges, negative for a virtual image on the object side; R is positive when the centre of curvature lies on the outgoing (emerging) side, i.e. a surface convex towards the incident light.
- Refraction at a spherical surface obeys Snell's law $n_1 \sin i = n_2 \sin r$ at each point; for paraxial rays this leads to the surface equation below.
- **Apparent depth**: an object in a denser medium appears closer to the surface when viewed normally, because rays bend away from the normal on leaving it.
- A plane surface has $R \to \infty$, so the term $\frac{n_2 - n_1}{R}$ vanishes; then $\frac{n_1}{u} + \frac{n_2}{v} = 0$ and $\frac{v}{u} = -\frac{n_2}{n_1}$.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Refraction at a spherical surface	$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$	n_1 = refractive index of the medium containing the object; n_2 = refractive index of the other medium; u = object distance; v = image distance; R = radius of curvature (m)
Apparent depth	Apparent depth = real depth × $\frac{\text{n of observer's medium}}{\text{n of object's medium}}$; water viewed from air: apparent depth = real depth/n	Depths in m; n = refractive index of water for the water-from-air case
Second focal length	$f_2 = \frac{n_2 R}{n_2 - n_1}$	f_2 = second focal length (m), obtained from $u \to \infty$; n_1, n_2 = refractive indices; R = radius of curvature (m)
First focal length	$f_1 = \frac{n_1 R}{n_2 - n_1}$	f_1 = first focal length (m), obtained from $v \to \infty$; n_1, n_2 = refractive indices; R = radius of curvature (m)
Lateral magnification	$m = -\frac{n_1 v}{n_2 u}$	m = lateral magnification (no unit); n_1, n_2 = refractive indices; u = object distance; v = image distance

TIPS TO ANSWER

- State which medium is n_1 (object side) and which is n_2, then give the sign of R with a reason (centre on outgoing side: positive).
- Substitute with consistent units, solve for $\frac{1}{v}$ and then v; the sign of v shows whether the image is real (outgoing side) or virtual (object side).
- Quote the position relative to the surface: 'v cm inside the glass' or 'v cm on the object side'.
- For plane surfaces, use the apparent depth relation and give the answer in the same unit as the real depth.

SECTION A TRAPS

Trap: Interchanging n_1 and n_2, so that the numerator $(n_2 - n_1)$ changes sign and v is wrong.

Trap: Using the wrong sign of R for a concave surface (R negative when the centre of curvature is on the incident side).

Trap: Applying apparent depth = real depth × n. It is divided by n when looking from air into the denser medium.

Trap: Treating the refracting surface as a thin lens; the surface equation involves two indices and a single radius.

TRICKS

Trick: Check the sign of v by physical reasoning: from a less dense medium into a denser one through a convex surface, the rays converge more, so v is positive and shorter than in the plane case.

Trick: Plane surface test: $v = -\frac{n_2}{n_1}u$, so for a fish in water viewed from air v is shorter than u.

23.3 Thin Lenses

DEFINITIONS & LAWS

- **Sign convention (real-is-positive):** u positive for a real object; v positive for a real image (on the opposite side of the lens), negative for a virtual image; f positive for a converging (convex) lens and negative for a diverging (concave) lens.
- The **principal focus** of a converging lens is the point on the axis to which rays parallel and close to the axis converge after refraction through the lens.
- The **power** of a lens is the reciprocal of its focal length in metres, $P = \frac{1}{f}$, with unit dioptre (D).
- **Linear magnification** $m = \frac{h_i}{h_o} = -\frac{v}{u}$; a real image from a single lens is inverted (m negative); a virtual image is upright (m positive).
- A convex lens forms a real, inverted image when u > f and a virtual, upright, magnified image when u < f (magnifying glass); a concave lens always forms a virtual, upright, diminished image.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Thin lens equation	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	u = object distance; v = image distance; f = focal length (all distances in the same unit)
Lensmaker's formula	$\frac{1}{f} = (n - 1)(\frac{1}{R_1} + \frac{1}{R_2})$	n = refractive index of lens material; R_1, R_2 = radii of the two surfaces, each positive when its centre of curvature lies on the side from which the light emerges (biconvex: both positive); if the form $\frac{1}{R_1} - \frac{1}{R_2}$ is used, R_2 is negative for a biconvex lens
Linear magnification	$m = -\frac{v}{u}$	m = magnification (no unit); u = object distance; v = image distance
Power of a lens	$P = \frac{1}{f}$	P = power (dioptre, D); f = focal length (m)
Lenses in contact	$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$; $P = P_1 + P_2$	f = combined focal length; f_1, f_2 = focal lengths of the lenses; P = combined power (D); P_1, P_2 = powers of the lenses (D)
Real image same size as object	$u = v = 2f$	u = object distance; v = image distance; f = focal length
Minimum object-image distance	$u + v = 4f$	Minimum distance between object and a real image is 4f; f = focal length

TIPS TO ANSWER

- State the sign convention, write f with the correct sign, and find v from $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$; then give m and describe the image (real/virtual, inverted/upright, magnified/diminished).
- For combinations in contact, add the powers or reciprocal focal lengths and show the working to 3 s.f.
- Convert f to metres before calculating power; give the unit D.
- For ray diagrams, use two principal rays (parallel to axis then through F; through the optical centre undeviated), arrows, and label F and the optical centre.
- The lensmaker's equation is written with $(\frac{1}{r_1} + \frac{1}{r_2})$ or $(\frac{1}{r_1} - \frac{1}{r_2})$; each form has its own sign convention. Either is accepted unless the question asks you to show a particular form. Then derive that form with its own convention: in the $(\frac{1}{r_1} - \frac{1}{r_2})$ form, r is positive when the centre of curvature is on the right (outgoing) side of the surface, and the image from the first surface is a virtual object for the second.

SECTION A TRAPS

Trap: Calculating power with f in cm (a 25 cm lens is 4.0 D, not 0.040 D).

Trap: Adding focal lengths instead of reciprocals for two lenses in contact: the combination of 25 cm and 40 cm is 15.4 cm, not 65 cm.

Trap: Using the lensmaker's formula with the sign convention for one radius reversed; check that a biconvex lens in air gives a positive f.

Trap: Quoting the sign of v the wrong way round: in real-is-positive a negative v is a virtual image on the object side.

TRICKS

Trick: Unit check: P in D requires f in m; ×100 slip is the commonest error.

Trick: Magnification ratio method: a real image m times larger has v = mu, so $\frac{1}{f} = (1 + \frac{1}{m})/u$ and $u = f(1 + \frac{1}{m})$.